Assessment Tool - 1

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1. Scenario: Event Communication Security

A student organization is preparing to share confid-ential event details with its core members using a simple classical encryption method. During testing a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

a) Reconstruct how the encrypted message might appear.

b) Explain how an authorized member, aware of the encoding rule, would retrieve the original message

Event Communication Security - Caesar Cipher

a) Encryption using a shift of 4.

Original message: MEET AT NOON

Each letter is shifted 4 positions forward in the alphabet

- M → Q, E → I, E → I, T → X
- A → E, T → X
- N → R, O → S, O → S, N → R

Encrypted message: QIIX EX RSSR

b) Decryption

The authorized member shifts each letter 4 positi-ons backward to recover the original message

Example: Q → M, I → E

original message: MEET AT NOON

2. Scenario: Intercepted Suspicious Message
A cybersecurity intern intercepts the coded message "GSRH RH Z HVXIVG" during network monitoring. The structure of the text message a simple substitution mechanism rather than complex encryption.
a) Deduce the most likely original message
b) Justify your answer by analyzing repeating patterns, letter structures, or known cipher characteristics

Intercepted Suspicious Message - Atbash Cipher
a. Original message
Encrypted message: GSRH RH Z HVXIVG
Using the Atbash cipher, which reverses the alphabet (A ↔ Z, B ↔ Y, etc), the original message is

THIS IS A SECRET

b) Justification
- Each letter is replaced by its opposite letter in the alphabet.
- The word patterns match common English words. For example, G → T, S → H, R → I, and H → S, forming "THIS".
- The same substitution rule correctly decodes the entire message.

3) Scenario: Secure Messaging System

A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentially

a) Demonstrate how the encryption process transforms the message step-by-step.

b) Explain how the repeating keyword influences each character of the ciphertext.

a) Encryption of HELLO using key

Repeat the keyword until it matches the message length

Plain text	H	E	L	L	O
keyword	K	E	Y	K	E
Cipher text	R	I	J	V	S

using the rule $C = (P+K) \bmod 26$

where A = 0:

- H + K = R
- E + E = I
- L + Y = J
- L + K = V
- O + E = S

Encrypted message: RIJVS

b) Effect of repeating keyword.

The keyword determines the shift applied to each plaintext letter. Repeating KEY produces different shift across the message, making the cipher more complex than a single fixed-shift Caesar cipher

4. Scenario: Suspicious Digital Asset Investigation

During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

a) Analyze how investigators could detect the presence of hidden data within the image.

b) Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction

a) Detecting hidden data

Investigators can use the following techniques:

- Visual inspection: Look for unusual colors, Pixel Patterns, or image disortions
- File analysis: Examine file size, metadata, and file structure for suspicious information.
- Statistical analysis: Check Pixel values and least significant bits for unusual Patterns.
- steganalysis tools: Use forensic software to identify possible hidden messages or embedded files.

b. Two security enhancement techniques.

1. Encrypt the hidden message: Apply Strong encryption such as AES before embedded the data

2. Use randomized embedding: Distribute encrypted data across selected Pixels using a secret key, making detection more difficult

5. Scenario: Legacy Encryption in corporate Communication
A legacy system in a company still uses a transposition -based encryption method using three levels of arr -angement. A received message appears as "WECRLTEER DSOEEFECOAIVDEN".
a) Determine the most Portable original message by reversing the encryption process.
b) Critically evaluate why such a method is vulnerable in modern communication systems.
c) propose one practical enhancement to improve the confidentiality of message in this Scenario.

Legacy Encryption - Three-level Transposition Cipher

a) Recovering the original message

Ciphertext: WECRLTEERDSOEEFEAOCAIVDEN

The most Probable original message is

WE ARE DISCOVERED FLEE AT ONCE

Removing Spaces gives WEAREDISCOVEREDFLEEATONCE. The ciphertext is a known example of a rail-fence transposition cipher with three rails, where letters are rearranged in a zigzag Pattern.

b) why is it Vulnerable

.It only rearrange letters, it does not change them
.letter frequencies and repeated Patterns remains visible.

C. Practical enhancement.

Replace the transposition cipher with a modern authenticated encrypted algorithm such as AES-GCM, which protects message confidentiality and helps detect authorized modification